

Buddha Nath Sharma

Senior Research Fellow · Department of Physics · National Institute of Technology Sikkim

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EDUCATION

Ph.D. in Physics

National Institute of Technology Sikkim August 2023 – Present

- Thesis: *Applications of Topological Data Analysis in Nonlinear Dynamics*

M.Sc. Physics

NBBGC Tadong, Central University of Sikkim June 2019 – June 2021

B.Sc. (Hons.) Physics

NBBGC Tadong, Central University of Sikkim June 2016 – June 2019

COMPETITIVE EXAMINATIONS

Graduate Aptitude Test in Engineering (GATE) 2023

- Qualified GATE 2023 in Physics.

State Level Eligibility Test (SLET) 2022

- Qualified SLET 2022; eligible for appointment as Assistant Professor.

LANGUAGES & SKILLS

English Proficient

Hindi Fluent

Nepali Native

Technical Python · $\text{\LaTeX}$ · Topological Data Analysis · Nonlinear Dynamics · Financial Data Analysis

RESEARCH INTERESTS

My research interests involve the fields of topological data analysis and Nonlinear dynamics.

PROJECTS

Case Study on Drinking Water Supply Management System

BIS Student Chapter, Department of Basic Science, NIT Sikkim

- Aligned with IS 17482:2020 – Requirements for Piped Drinking Water Supply Service.
- Involved field visits to PHE divisions across multiple districts in Sikkim.

MENTORSHIPS

Undergraduate Mentees

Joe Opitz and Adam Moser

Montana Technological University, Montana, USA

February 2026 – Present

Capstone project title: *Mixup Barcodes for Topology-Aware Financial Decision Making*

WORKSHOPS AND CONFERENCES

International Conference on Complex Dynamical Systems & Applications

ISI Kolkata January 2024 Digha, West Bengal

Workshop on Topological Data Analysis

Sikkim University March 2024 Gangtok, Sikkim

Conference on Nonlinear Systems and Dynamics (CNSD)

Bharathidasan University March 2025 Trichy, Tamil Nadu

International Conference on Nonlinear Systems and Dynamics (CNSD)

IISER Thiruvananthapuram July 2026 Kerala

PUBLICATIONS

Journal Articles

Causality Analysis of COVID-19 Induced Crashes in Stock and Commodity Markets: A Topological Perspective

Buddha Nath Sharma, Anish Rai, S. R. Luwang, Md Nurujjaman, Sushovan Majhi

International Journal of Modern Physics C, 2025

Identifying Extreme Events in the Stock Market: A Topological Data Analysis

Anish Rai, Buddha Nath Sharma, S. R. Luwang, Md Nurujjaman, Sushovan Majhi

Chaos: An Interdisciplinary Journal of Nonlinear Science, 2024

Preprints

Interpretable Classification of Time Series Using Euler Characteristic Surfaces

Buddha Nath Sharma, Vishal Mandal, S. R. Luwang, Md Nurujjaman, Atish J Mitra, Sushovan Majhi

Research square preprint, April 2026

Regime Discovery and Intra-Regime Return Dynamics in Global Equity Markets

S. R. Luwang, Buddha Nath Sharma, Kundan Mukhia, Md Nurujjaman, Anish Rai, Filippo Petroni

arXiv preprint, 2026

Intraday Limit Order Price Change Transition Dynamics Across Market Capitalizations Through Markov Analysis

S. R. Luwang, Kundan Mukhia, Buddha Nath Sharma, Md Nurujjaman, Anish Rai, Filippo Petroni

arXiv preprint, 2026

Early-Warning Signals of Political Risk in Stablecoin Markets: Human and Algorithmic Behavior Around the 2024 US Election

Kundan Mukhia, Buddha Nath Sharma, S. R. Luwang, Md Nurujjaman, Chittaranjan Hens, Suman Saha, Tanujit Chakraborty

arXiv preprint, 2026

Detecting Regime Transitions in Dynamical Systems via the Mixup Euler Characteristic Profile

Sushovan Majhi, Atish Mitra, Santanu Nandi, Md Nurujjaman, Buddha Nath Sharma

arXiv preprint, 2026

REFEREES

Dr. Md. Nurujjaman

Associate Professor, Dept. of Physics
National Institute of Technology Sikkim
md.nurujjaman@nitsikkim.ac.in

Dr. Sushovan Majhi

Assistant Professor, Data Science Program
George Washington University, USA
s.majhi@gwu.edu

Prof. Atish J. Mitra

Professor, Dept. of Mathematical Sciences
Montana State University, USA
amitra@mtech.edu